Kun Qian

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Education

Huazhong University of Science and Technology **Wuhan, China**

B.Eng. in Automation 09.2022-current

- Average grade: 91.32/100 (5/166 | Top 3%)
- Relevant courses: Principles of automatic control (95), Sensor Technology (93), Digital Circuit and Logic Design (96), Computer Control Technology (90), Pattern Recognition and Machine Learning (95)

Publication

- **K. Qian**, Y.-L. Li, X.-K. Liu. Quantized Distributed Dual Algorithm for Resource Allocation with Gradient Tracking over Unbalanced Networks. *Chinese Control Conference, CCC 2025*. (Published | EI)
- Y.-L. Li, **K. Qian**, X.-K. Liu, Y.-W. Wang, P. Siano. Communication-Efficient Distributed Resource Allocation via Implicit Dual Gradient Tracking. *IFAC World Congress 2026*. (Under Review | EI)

Research Experience

National College Student Innovation Training Program 01.2024-04.2025

Lightweight Design and Implementation of Elastic Distributed Optimization Algorithms

- Designed a **fully distributed multi-agent physical platform** based on Raspberry Pi 4B to **verify real-time optimization algorithm**, supporting 3+ agents with synchronous communication for real-time distributed optimization tasks.
- Built a **distributed algorithm simulation platform**, incorporating modules like network topology, algorithm and quantizer, supporting **convenient comparison** of algorithms under different experimental scenarios.
- Proposed a **quantized dual gradient tracking method** to address **limited bandwidth** and unbalanced communication topologies, achieving linear convergence with an **accuracy of 10^{-10}** even at a **quantization level of 3**.
- Resulted in a first-author paper accepted by **CCC 2025.**

Research on distributed optimization under imperfect communication 11.2024-12.2024

Communication-Efficient Distributed Resource Allocation via Implicit Dual Gradient Tracking

- **Conducted comparative experiments** of distributed optimization algorithms via MATLAB simulations and Raspberry Pi physical platforms.
- Authored the experimental section, detailing methodology, implementation, and results analysis.

Competition Experience

- National Intelligent Car Competition for College Students, Outdoor Unmanned Vehicle (ROS-based) Group (09.2024-12.2024 | Team Leader)
  - Led a 5-member team to design motion control and path planning algorithms.
  - Developed control code using **PID control** in Python.
  - Developed a simplified path planning method combining **coordinate transformation** and PID control, reducing task completion time by around **20 seconds** compared to the default ROS global-local planning approach.
  - Won the **first prize** in the national finals. (among 493 teams nationwide)

Awards

- **National Scholarship** (2024 | Top 2%)
- **Outstanding Undergraduates in Terms of Academic Performance** (2024 | Top 1%)
- **National First Prize**, National Intelligent Car Competition for College Students (2024)

- **Merit Student of Huazhong University of Science and Technology** (2024)
- **Honorable Mention,** Mathematical Contest in Modeling (MCM) (2023)
- **First Prize (Provincial Level)**, National College Mathematics Competition for Non-Mathematics Majors (2023)

Skills

- **Language**: English (IELTS-6.5); Chinese (native)
- **Programming**: Python (ROS, NumPy), C(STM32), MATLAB (MCM, Simulink)
- **Tool**: LaTeX, Keil
- **Hardware:** Raspberry Pi, STM32 Microcontrollers

Activities

- Volunteer Tutor, Spring Charity Tutoring Program, teaching math to an underprivileged student weekly 02.2024-06.2024
- "Hundreds of Villages and Thousands of Households" Summer Social Practice 07.2023-08.2023
  - Investigated 15 randomly selected farmers, as well as the affiliated village committee and township government.
  - Formed an investigation report based on the investigation results.
- First Prize, School Badminton Skills Competition 05.2023

Referees

- **Xiaokang Liu**, Associate Professor, School of Artificial Intelligence and Automation
  Huazhong University of Science and Technology, xiaokangliu@hust.edu.cn
- **Min Ouyang**, Professor, School of Artificial Intelligence and Automation
  Huazhong University of Science and Technology, min.ouyang@hust.edu.cn
- **Cheng Cheng,** Associate Professor, School of Artificial Intelligence and Automation
  Huazhong University of Science and Technology, c_cheng@hust.edu.cn